

C1—From prior research in Element A, here are the design requirements.

1. Must be built within the 10 days provided.
2. Must be safe for teachers and students to use during class activities to prevent any kind of unintentional physical injuries.
3. Must not block either side's laser or the front display while the inclinometer is being used.
4. Must hold the inclinometer in a fixed position after adjustment so that repeated measurements are stable and consistent.
5. Must be adjustable in both height and angle so that users can make the measurement shown on the screen with a range (0° , 90°).
6. Must remain stable on a flat classroom or lab surface and must not tip over during normal use.
7. Must be durable enough to withstand repeated classroom use without loosening, breaking, or losing alignment for at least 10 years.
8. Must be easily adjustable by human hands.
9. Must be light enough for one person to move and reposition easily.
10. Must fit within an 18-inch storage space.
11. Should include Richard Montgomery logo/design on the side.

C2—Rank the requirement by priority.

C3—High or low priority?

<i>Requirement Number</i>	<i>Priority (High or Low)</i>	<i>The Design Requirement</i>
1	Highest	Must be built within the 10 days provided.
2	Highest	Must be safe for teachers and students to use during class activities to prevent any kind of unintentional physical injuries.
3	Highest	Must not block either side's laser or the front display while the inclinometer is being used.
4	High	Must hold the inclinometer in a fixed position after adjustment so that repeated measurements are stable and consistent.
5	High	Must be adjustable in both height and angle so that users can make the measurement shown on the screen with a range (0° , 90°).
6	High	Must remain stable on a flat classroom or lab surface and must not tip over during normal use.
7	High	Must be durable enough to withstand repeated classroom use without loosening, breaking, or losing alignment for at least 10 years.
8	High	Must be easily adjustable by human hands.
9	High	Must be light enough for one person to move and reposition easily.
10	Medium	Must fit within an 18-inch storage space.
11	Medium	Should include Richard Montgomery logo/design on the side

C4—Measurable and Objective Data

C5—Your description is actually able to be built.

<i>Design Requirement №</i>	<i>Data that we can test with this requirement</i>
1	Prototype is completed within 10 calendar days, shown by build log, dated photos, or a timeline of work sessions.
2	0 injuries in 10 setup/use experiments, no sharp exposed edges, and no pinch points that catch fingers during adjustment.
3	In 10 out of 10 tests, both lasers project clearly onto the target surface and the front screen remains fully visible while the inclinometer is mounted.
4	After adjustment and leaving it untouched for 1 minute, the displayed angle changes by no more than 1°; when measuring the same target 5 times, the readings stay within a $\pm 2^\circ$ accuracy.
5	The stand can be adjusted to at least 5 different heights and can tilt from 0° to 90°, verified with an angle measurement tool or the inclinometer itself.
6	On a flat surface, the stand remains upright for 10 out of 10 tests, and during normal adjustment, it does not tip and does not slide more than 1 inch.
7	After 620 setup/use cycles or another repeated-cycle test based on class use, the stand shows no broken parts, no loose joints, and no angle drift greater than 1°.
8	A user can complete one height or angle adjustment using only their hands, with no tools, in under 15 seconds in 5 out of 5 tests.
9	The stand has a mass of 5 pounds or less, and one person can carry and reposition it from one table to another in one trip.
10	When fully folded or collapsed, the stand's largest dimension is 18 inches or less.
11	Logo/design is visibly present on at least one side of the final prototype and is at least 2" $\times$ 2" in size.

C6—Stakeholder approval

Do you have any other thoughts or suggestions regarding the design, restrictions, or requirements of our inclinometer stand?

2 responses

Your priority of specifications looks good. Will I get all these for the price of one case of airheads? This is the time that you can renegotiate if needed.

Nice to see the range of adjustments, ut what I really want to see is the range of angles that can be measured connecting two dots (one on the wall and one on the floor)

All the best, Dr K
karl.f.kovacs@mcpsmd.net

Google form:

<https://docs.google.com/forms/d/e/1FAIpQLSfyXim0mtE3wNJ6-gzfHWFRlGjVfoC9rjhmbBkNOIGItkl1LQ/viewform?usp=dialog>

We didn't end up finding Mrs Harmon despite sending the form to her.

<i>Design Requirement №</i>	<i>Stakeholder Approval</i>
1	Reviewed, revised if needed, and approved by Dr. Kovacs on April 21st. He confirmed this requirement and its priority level matched his needs for the inclinometer stand.
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